# SHEPETIVKA, Ukraine

WMO: **333170**

Lat: **50.1614N** Lon: **27.0369E** Elev: **278** StdP: **98.03** Time Zone: **2.00 (EUE)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-18.3** | **-14.9** | **-20.4** | **0.6** | **-17.8** | **-17.4** | **0.8** | **-14.5** | **14.7** | **0.4** | **10.5** | **-0.9** | **2.1** | **90** | **0.505** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **10.0** | **29.7** | **21.0** | **28.1** | **19.9** | **26.6** | **19.0** | **21.9** | **27.8** | **20.9** | **26.5** | **19.9** | **25.0** | **2.8** | **180** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **19.9** | **15.1** | **24.8** | **19.0** | **14.3** | **23.5** | **18.1** | **13.5** | **22.3** | **65.2** | **27.7** | **61.6** | **26.5** | **58.3** | **25.1** | **25.9** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **9.7** | **7.4** | **6.2** | DB | **-21.6** | **32.1** | **3.6** | **1.7** | **-24.1** | **33.3** | **-26.2** | **34.3** | **-28.2** | **35.2** | **-30.8** | **36.4** | (4) |
| (5) | | | | WB | **-21.8** | **23.3** | **3.5** | **1.4** | **-24.3** | **24.3** | **-26.4** | **25.1** | **-28.4** | **25.8** | **-30.9** | **26.8** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **8.2** | **-3.9** | **-2.6** | **1.9** | **9.1** | **14.6** | **17.6** | **19.4** | **18.7** | **13.8** | **8.2** | **2.9** | **-2.1** | (6) |
| (7) | | DBStd | | **9.46** | **5.62** | **5.63** | **4.77** | **4.32** | **4.07** | **3.44** | **3.12** | **3.35** | **3.92** | **4.40** | **4.76** | **5.20** | (7) |
| (8) | | HDD10.0 | | **1802** | **431** | **353** | **253** | **66** | **8** | **0** | **0** | **0** | **10** | **88** | **218** | **375** | (8) |
| (9) | | HDD18.3 | | **3863** | **689** | **587** | **510** | **277** | **128** | **55** | **25** | **36** | **143** | **315** | **464** | **633** | (9) |
| (10) | | CDD10.0 | | **1142** | **0** | **0** | **1** | **39** | **150** | **229** | **292** | **271** | **125** | **32** | **3** | **0** | (10) |
| (11) | | CDD18.3 | | **161** | **0** | **0** | **0** | **1** | **12** | **33** | **59** | **48** | **8** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **1375** | **0** | **0** | **0** | **10** | **123** | **257** | **487** | **425** | **72** | **1** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **319** | **0** | **0** | **0** | **1** | **18** | **43** | **125** | **119** | **13** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **2.7** | **3.2** | **3.3** | **3.3** | **3.0** | **2.6** | **2.3** | **2.0** | **2.0** | **2.3** | **2.7** | **3.1** | **3.1** | (14) |
| (15) | Precipitation | PrecAvg | | **712** | **42** | **34** | **34** | **54** | **68** | **99** | **112** | **73** | **60** | **42** | **45** | **49** | (15) |
| (16) | | PrecMax | | **1038** | **127** | **68** | **83** | **120** | **193** | **244** | **232** | **209** | **184** | **122** | **140** | **92** | (16) |
| (17) | | PrecMin | | **428** | **8** | **3** | **0** | **7** | **22** | **12** | **10** | **12** | **7** | **3** | **9** | **4** | (17) |
| (18) | | PrecStd | | **139** | **27** | **17** | **23** | **32** | **35** | **54** | **51** | **40** | **41** | **30** | **27** | **21** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **7.5** | **10.1** | **16.3** | **24.6** | **28.8** | **29.7** | **31.7** | **32.2** | **28.3** | **22.5** | **15.9** | **9.2** | (19) |
| (20) | | | MCWB | | **5.8** | **7.4** | **10.3** | **15.4** | **19.0** | **21.1** | **22.1** | **21.8** | **18.5** | **15.8** | **11.8** | **7.7** | (20) |
| (21) | | 2% | DB | | **5.3** | **7.2** | **13.0** | **21.0** | **26.6** | **27.9** | **29.9** | **29.7** | **25.4** | **19.5** | **12.8** | **7.2** | (21) |
| (22) | | | MCWB | | **4.1** | **5.3** | **8.3** | **13.5** | **17.6** | **19.8** | **21.5** | **20.8** | **17.2** | **14.3** | **10.5** | **6.1** | (22) |
| (23) | | 5% | DB | | **3.7** | **5.3** | **10.6** | **18.7** | **24.4** | **26.3** | **27.9** | **27.7** | **22.9** | **16.6** | **10.9** | **5.5** | (23) |
| (24) | | | MCWB | | **2.8** | **4.0** | **6.9** | **12.2** | **16.6** | **19.0** | **20.3** | **19.6** | **16.5** | **12.7** | **9.4** | **4.6** | (24) |
| (25) | | 10% | DB | | **2.4** | **3.7** | **8.3** | **16.5** | **22.2** | **24.6** | **26.1** | **25.7** | **20.6** | **14.3** | **9.2** | **3.7** | (25) |
| (26) | | | MCWB | | **1.6** | **2.5** | **5.6** | **10.9** | **15.5** | **18.2** | **19.5** | **18.7** | **15.3** | **11.5** | **8.0** | **2.9** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **6.0** | **7.9** | **10.7** | **17.3** | **20.1** | **22.3** | **23.2** | **23.5** | **19.9** | **16.7** | **12.3** | **8.2** | (27) |
| (28) | | | MCDB | | **7.2** | **9.9** | **15.2** | **22.7** | **27.1** | **27.9** | **30.1** | **28.6** | **26.8** | **21.4** | **14.8** | **8.9** | (28) |
| (29) | | 2% | WB | | **4.3** | **5.6** | **8.7** | **14.1** | **18.5** | **21.0** | **22.0** | **21.6** | **18.2** | **14.7** | **10.9** | **6.2** | (29) |
| (30) | | | MCDB | | **5.1** | **7.1** | **12.3** | **19.8** | **24.7** | **26.4** | **27.9** | **27.7** | **23.1** | **18.3** | **12.6** | **7.0** | (30) |
| (31) | | 5% | WB | | **2.8** | **4.1** | **7.2** | **12.7** | **17.4** | **19.9** | **21.1** | **20.5** | **17.0** | **13.3** | **9.4** | **4.7** | (31) |
| (32) | | | MCDB | | **3.5** | **5.2** | **10.0** | **17.8** | **22.9** | **24.6** | **26.5** | **26.0** | **21.3** | **16.1** | **10.7** | **5.4** | (32) |
| (33) | | 10% | WB | | **1.7** | **2.6** | **5.9** | **11.5** | **16.2** | **18.8** | **20.1** | **19.4** | **15.8** | **11.8** | **8.1** | **3.1** | (33) |
| (34) | | | MCDB | | **2.2** | **3.5** | **8.1** | **15.5** | **20.9** | **23.3** | **25.0** | **24.3** | **19.7** | **13.9** | **9.2** | **3.7** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **4.5** | **5.3** | **6.9** | **9.4** | **10.4** | **9.9** | **10.0** | **10.4** | **9.0** | **7.7** | **4.9** | **4.2** | (35) |
| (36) | | 5% DB | MCDBR | | **4.2** | **6.5** | **10.6** | **12.8** | **13.3** | **12.0** | **12.7** | **13.1** | **12.8** | **11.2** | **8.1** | **5.3** | (36) |
| (37) | | | MCWBR | | **3.6** | **4.9** | **6.6** | **6.6** | **6.0** | **5.6** | **5.5** | **5.6** | **6.1** | **6.3** | **5.4** | **4.5** | (37) |
| (38) | | 5% WB | MCDBR | | **4.0** | **6.2** | **9.7** | **11.8** | **11.9** | **10.8** | **11.5** | **11.7** | **11.1** | **10.2** | **7.3** | **5.1** | (38) |
| (39) | | | MCWBR | | **3.6** | **4.9** | **6.4** | **6.4** | **5.9** | **5.7** | **5.4** | **5.6** | **6.0** | **6.3** | **5.1** | **4.4** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.308** | **0.339** | **0.376** | **0.429** | **0.407** | **0.403** | **0.434** | **0.432** | **0.396** | **0.360** | **0.340** | **0.306** | (40) |
| (41) | | taud | | | **2.221** | **2.171** | **2.188** | **2.129** | **2.265** | **2.295** | **2.221** | **2.222** | **2.278** | **2.360** | **2.318** | **2.279** | (41) |
| (42) | | Ebn at Noon | | | **706** | **776** | **816** | **811** | **851** | **856** | **820** | **800** | **789** | **753** | **667** | **655** | (42) |
| (43) | | Edh at Noon | | | **74** | **100** | **119** | **141** | **128** | **126** | **133** | **127** | **108** | **83** | **68** | **61** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **0.89** | **1.72** | **2.90** | **4.28** | **5.46** | **5.80** | **5.52** | **4.90** | **3.38** | **2.02** | **0.86** | **0.61** | (44) |
| (45) | | RadStd | | | **0.16** | **0.27** | **0.42** | **0.45** | **0.49** | **0.43** | **0.43** | **0.33** | **0.51** | **0.32** | **0.09** | **0.12** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **+0.65** | **N/A** | **N/A** | **+0.71** | **+0.60** | **N/A** | **N/A** | **-184** | **+104** | **+40** | (46) |
| (47) | Regional (7 neighbors) | **+0.71** | **N/A** | **N/A** | **+0.90** | **+0.56** | **+0.44** | **-139** | **-210** | **+123** | **+50** | (47) |

Nomenclature: See separate page